Advances in M87 Near-Horizon Theory

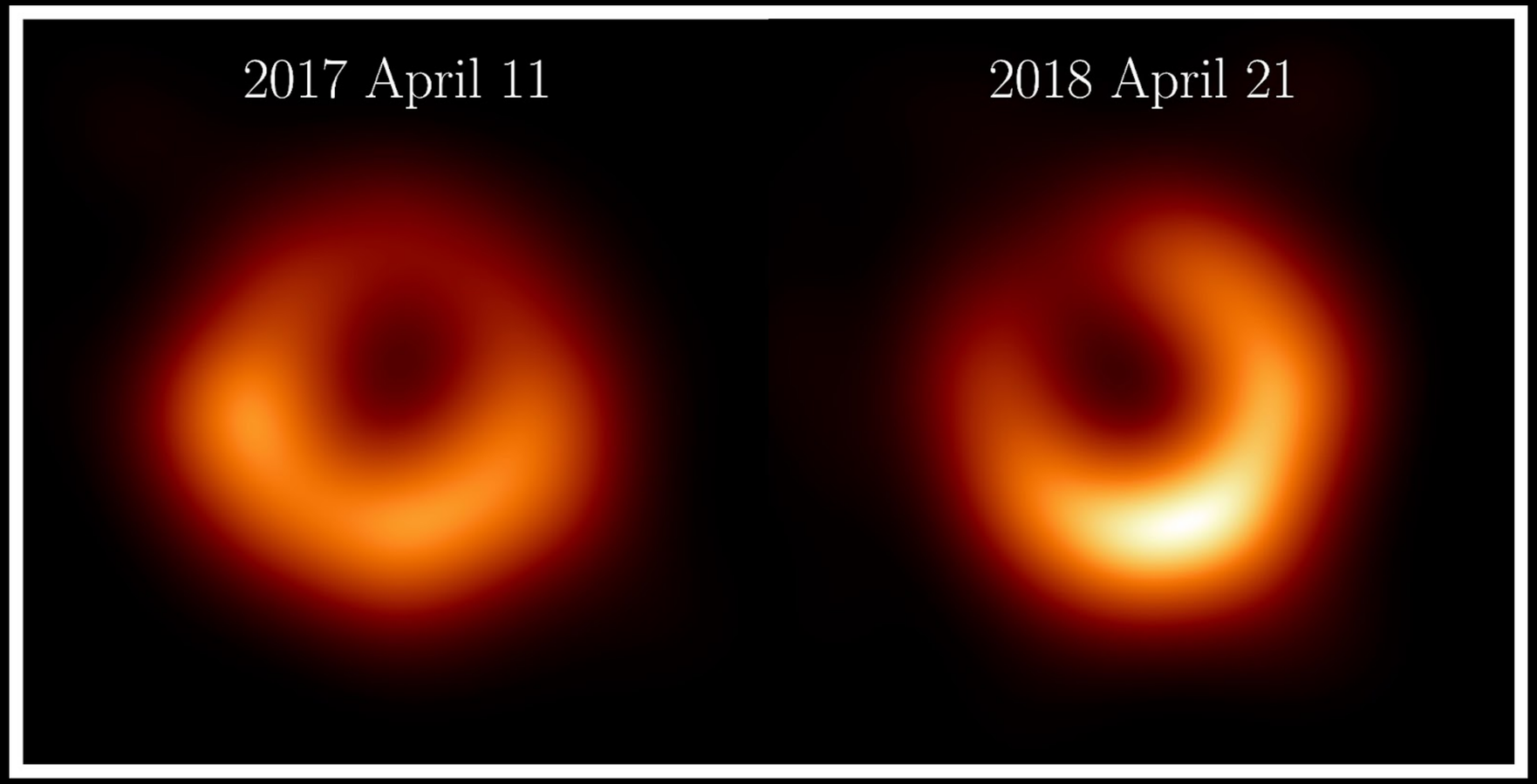
Koushik Chatterjee
U. of Maryland
(On behalf of the EHT collaboration)



Event Horizon Telescope



Multi-Year Observations of A Black Hole



EHT M87 2017 paper I (EHTC+19, ApJL)
EHT M87 2018 paper I (EHTC+24, A&A)

Koushik Chatterjee

AAS245

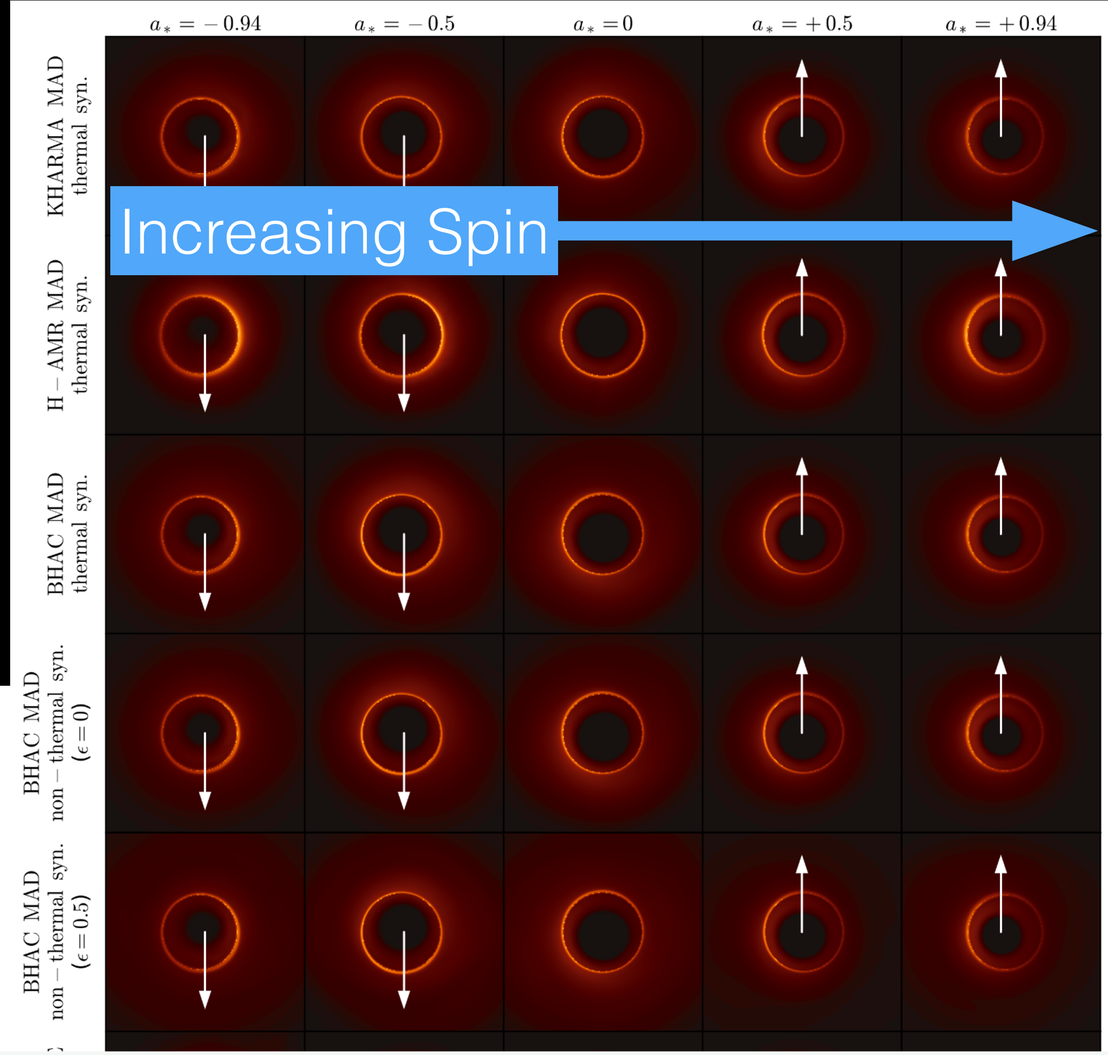
Building a New Image Library

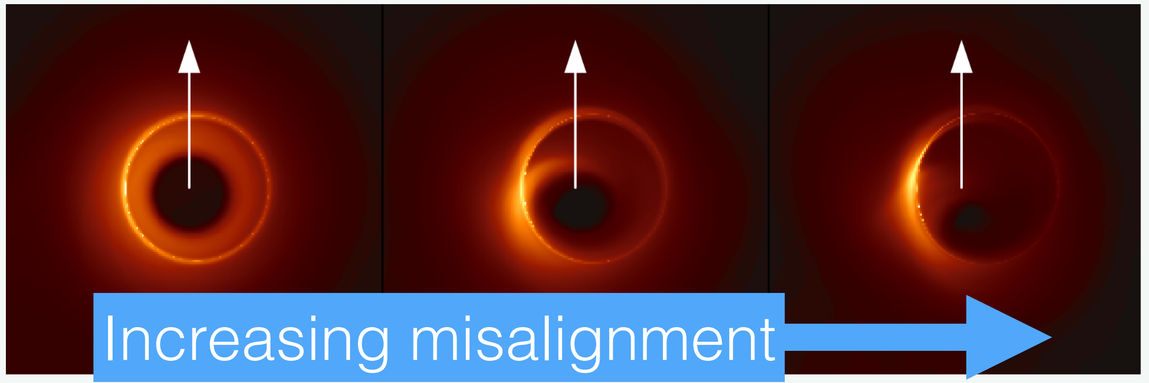
EHT M87 2018
paper II

~184,000 images including
- Thermal Synchrotron
- Models with Non-Thermal Synchrotron
- Misaligned accretion models
- Models with radiation physics

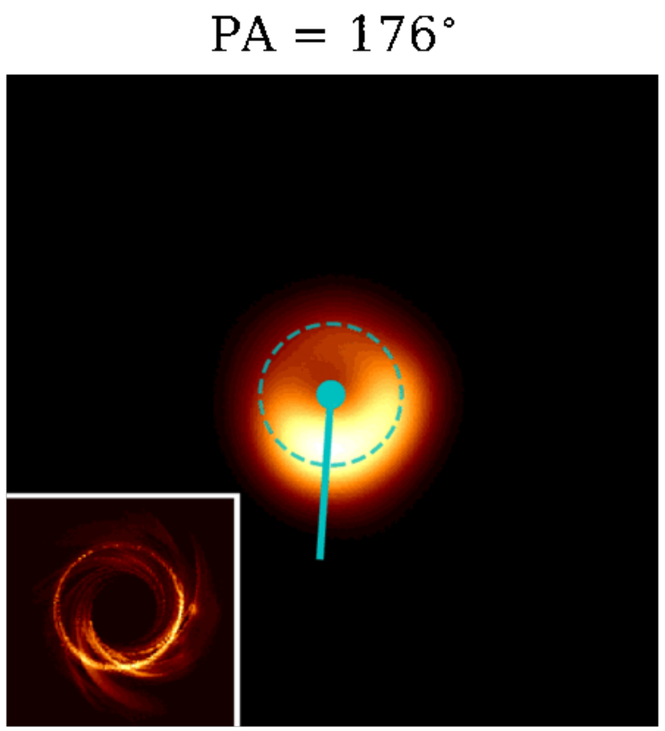
Table 2. Image model library

Code	RT Code	$R_{\rm high}$	$R_{\rm low}$	$\sigma_{\rm cut}$	Inclination	$\frac{\Delta t}{(10^3 GM/c^3)}$	Notes
			fiducial, thermal (synchrotron) models				
KHARMA	ipole	1,10,40,160	1,10	1	17°,163°	25–30	black hole spin axis are pointing away and toward Earth
H-AMR	BHOSS	1,10,20,40,80,160	1	1	17°,163°	25–30	black hole spin axis are pointing away and toward Earth
BHAC	BHOSS	40,80,160	1,10	3	17° or 163°	25–30	black hole spin axis are pointing away from Earth
			fiducial, nonthermal (synchrotron) models				
BHAC	BHOSS	40,80,160	1	3	17° or 163°	25–30	variable κ, ϵ = (0.0, 0.5), black hole spin axis are pointing away from Earth
			exploratory, tilted disk models				
H-AMR	BHOSS	1,10,20,40,80,160	1	1	163°	100-103	tilt angle = (0°, 30°, 60°), azimuth angle = (180°, 210°, 240°), black hole spin axis are pointing away from Earth
			exploratory, radiative models				
koral	grtrans	-	-	1	163°	11-16	T_e is directly from simulation, black hole spin axis are pointing away from Earth

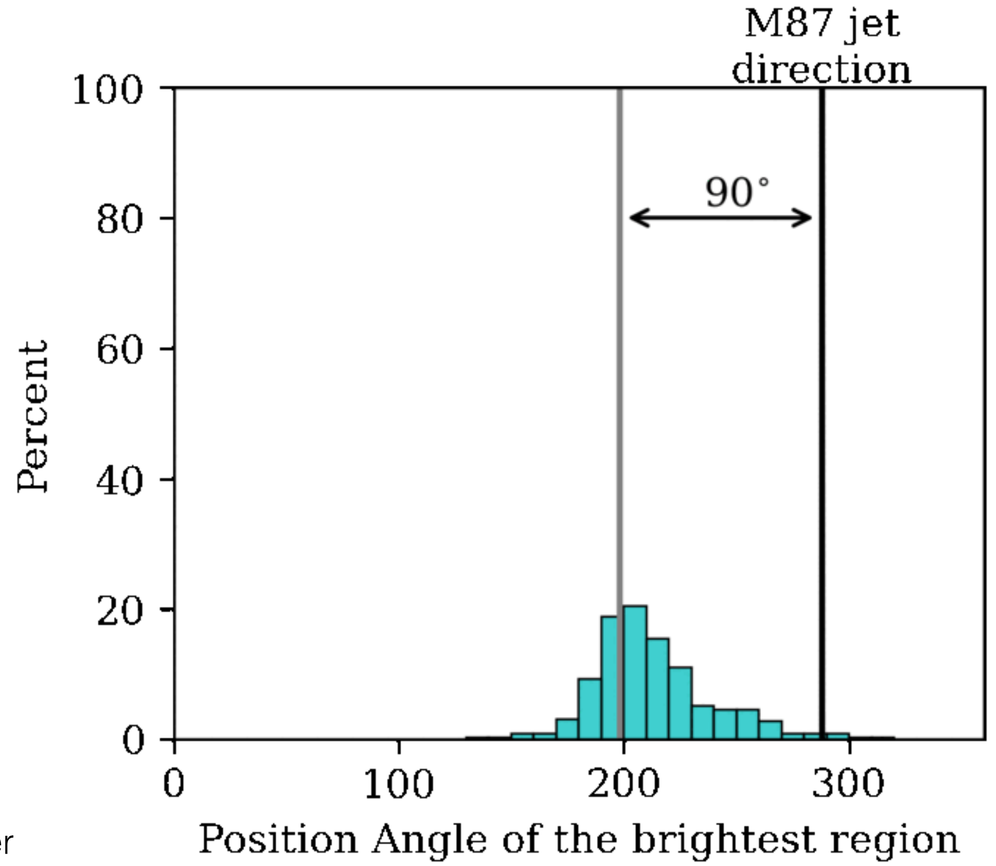




Multi-Year Analysis: Effects of Turbulence



visualization credit: Hung-Yi Pu
simulation credit: Abhishek Joshi & Ben Prather



The PA changes quite a lot over the duration of the simulation (4.5 years)

EHT M87 2018 paper II (EHTC+25, A&A, accepted)

2017 + 2018 M87 Results

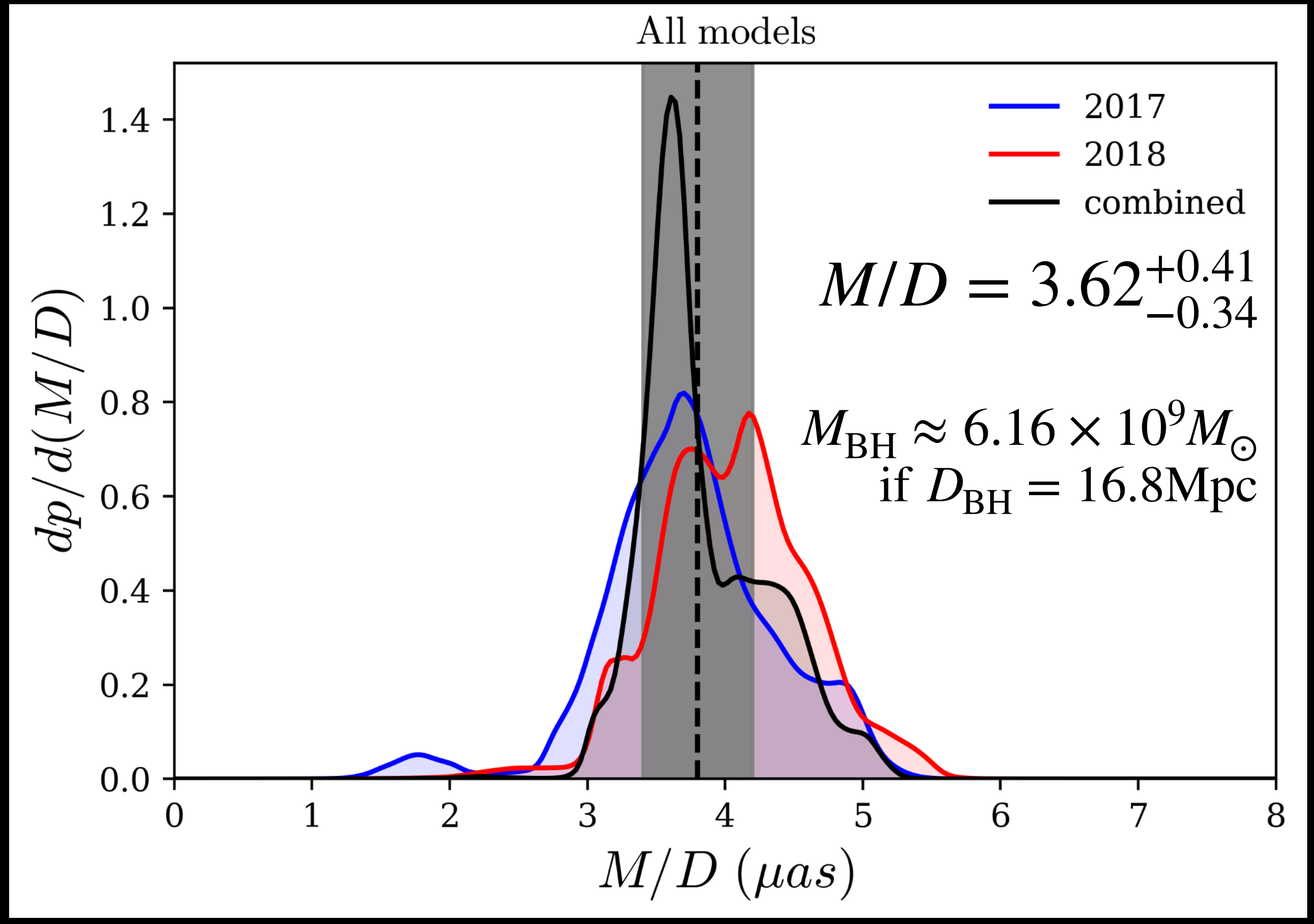
Bayesian multi-epoch scoring combines:

- Comparison of each image with the EHT observed Image
- Likelihood of consistency between an image set and the EHT data

Strongly magnetized models fair better while retrograde black hole spins show more variation in image structure

No evidence of misalignment (yet!)

- Open question about precession of the accretion flow and the jet!



EHT M87 2018 paper II
(EHTC+25, A&A, accepted)

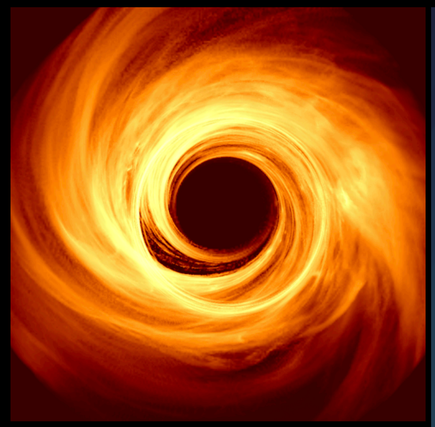
Theoretical Developments….

Magnetically Arrested Disks

- Magnetic eruptions & Flares
- Powerful jets
- Variability in emission
- Complex image structure

Igumenshchev+03, Narayan+03, Tchekhovskoy+11, McKinney++, Porth++, Begelman++, Chael++, Chatterjee++, Davelaar++, Dexter++, Liska++, Ressler++, Ricarte++, Ripperda++, White++ and many more

Quiescent

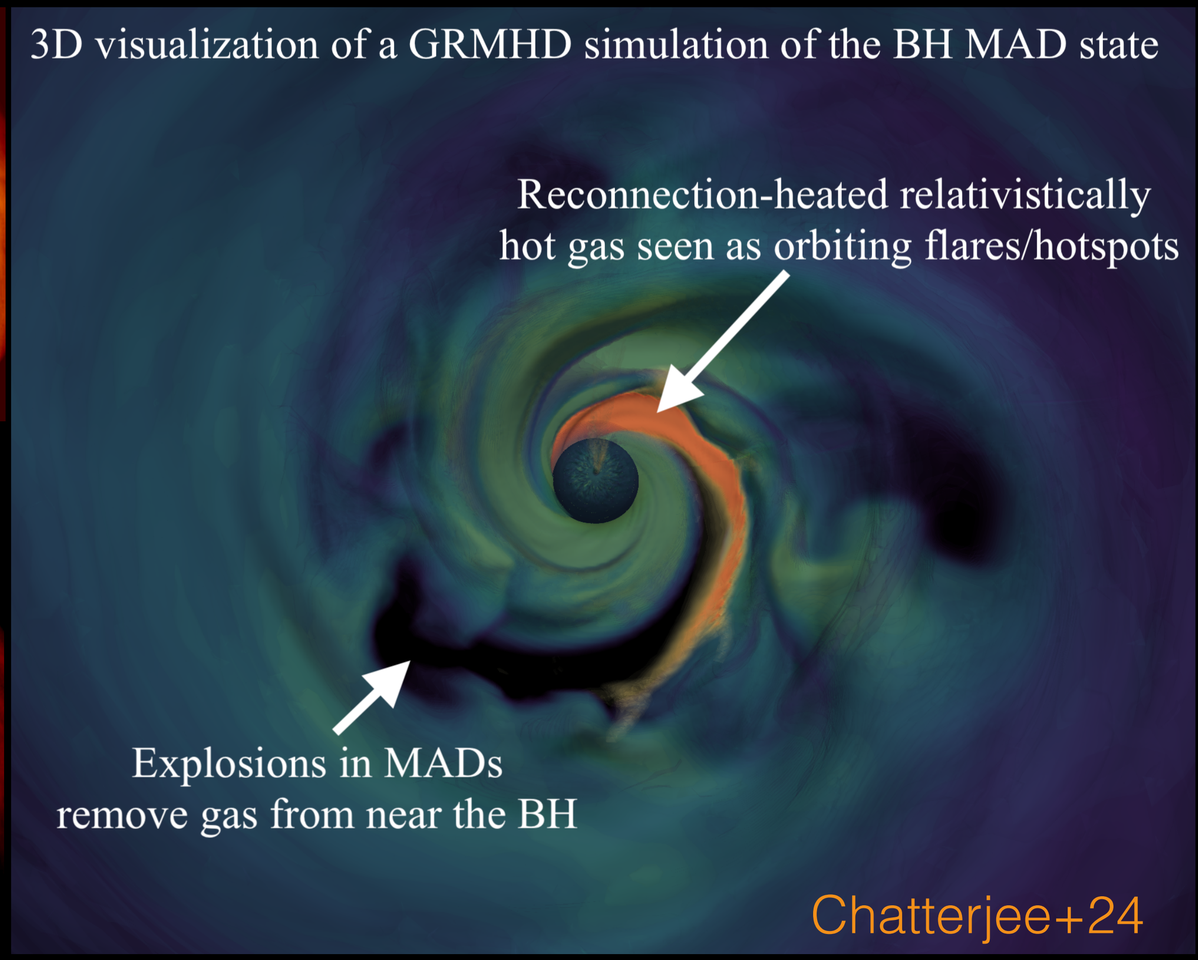


Flaring

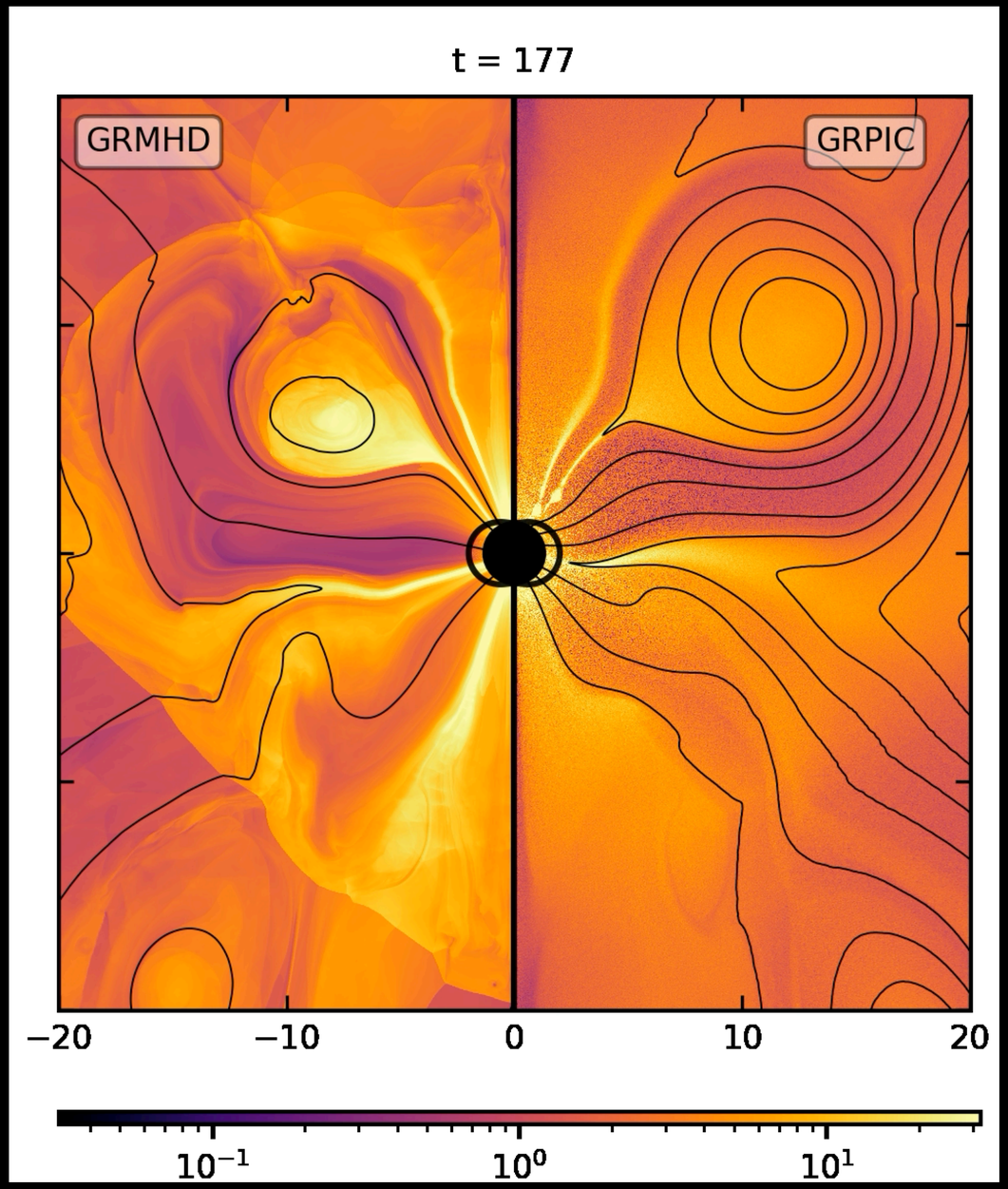


Gelles+22

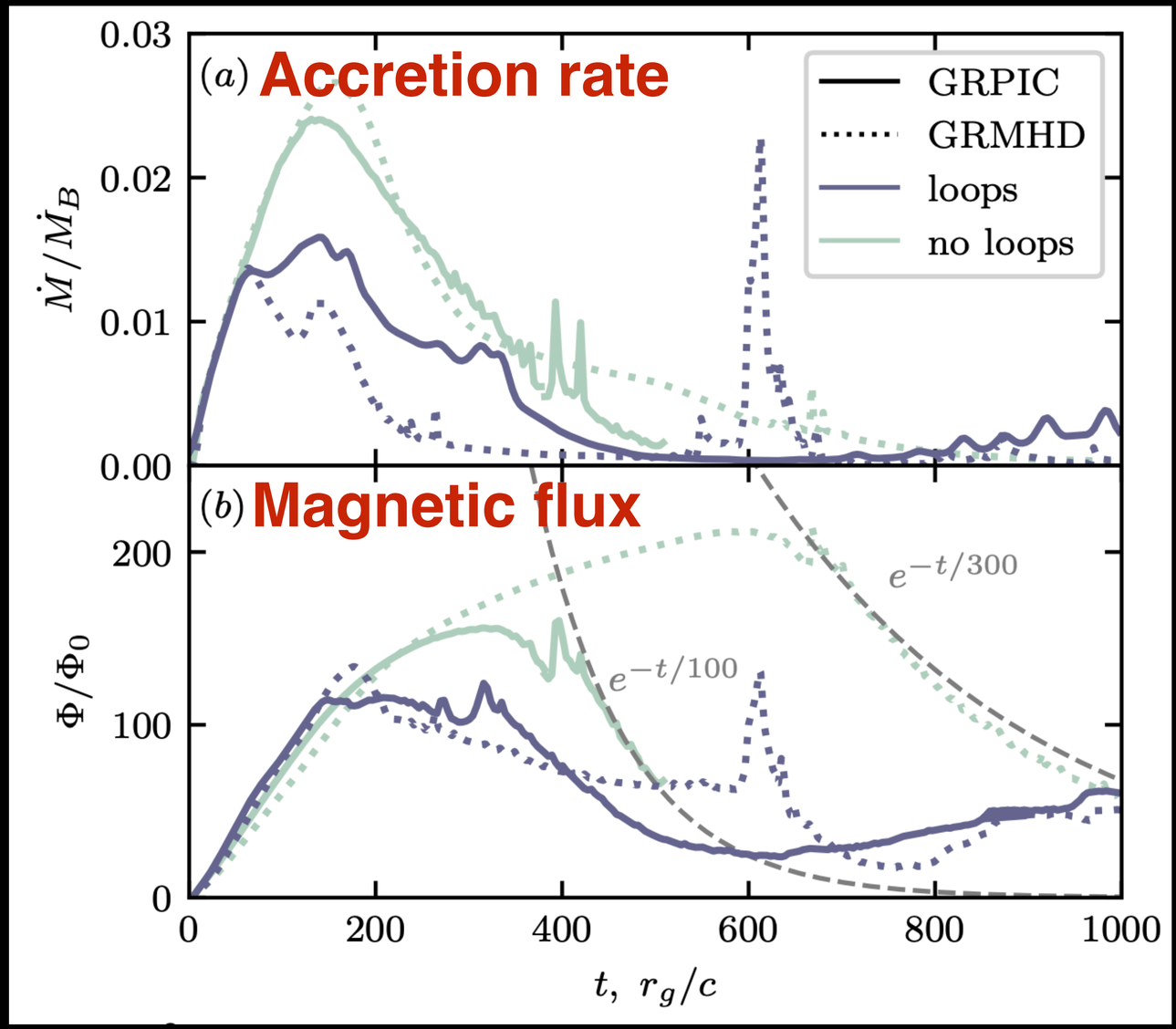
MADs in action



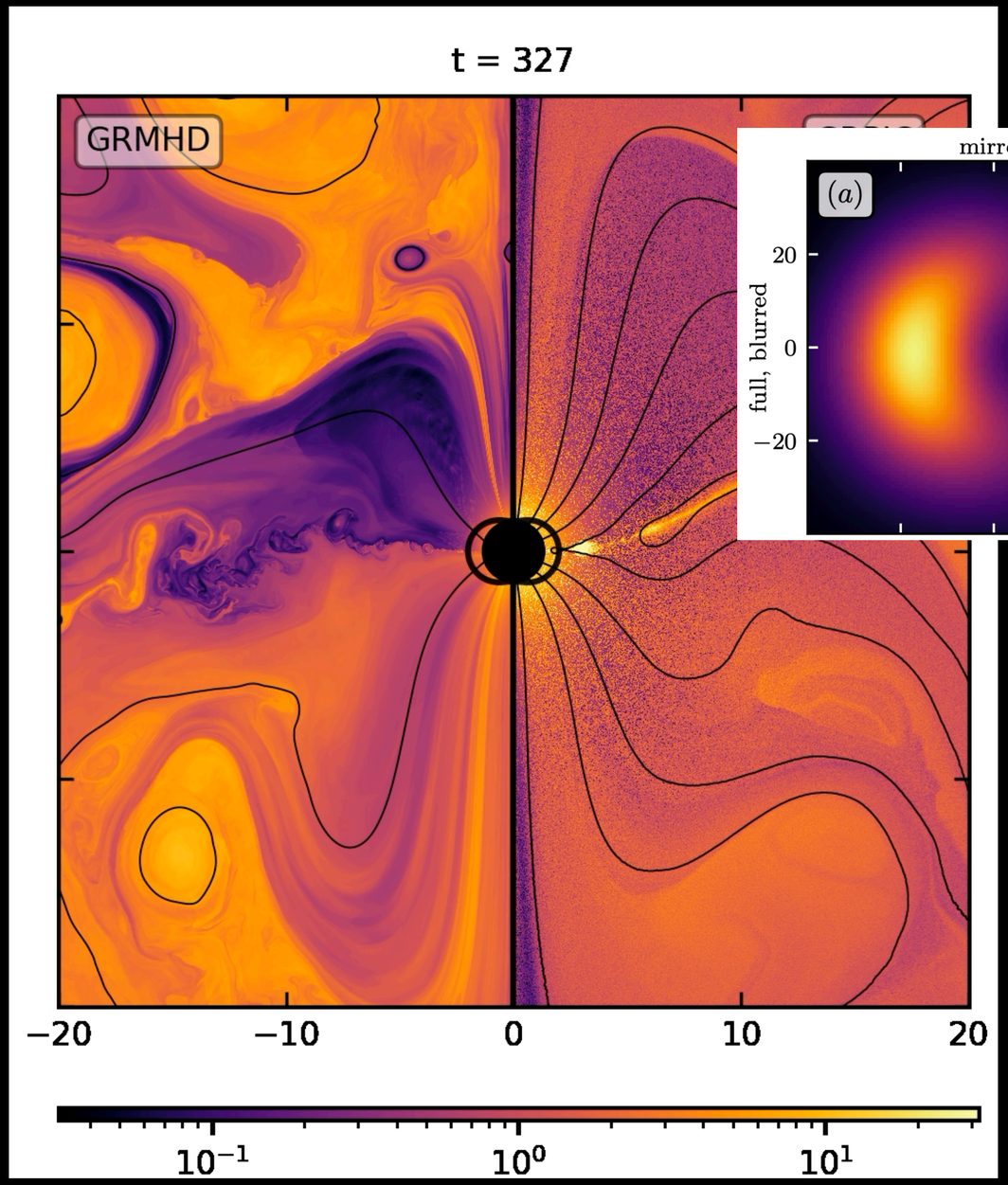
Theoretical Developments….

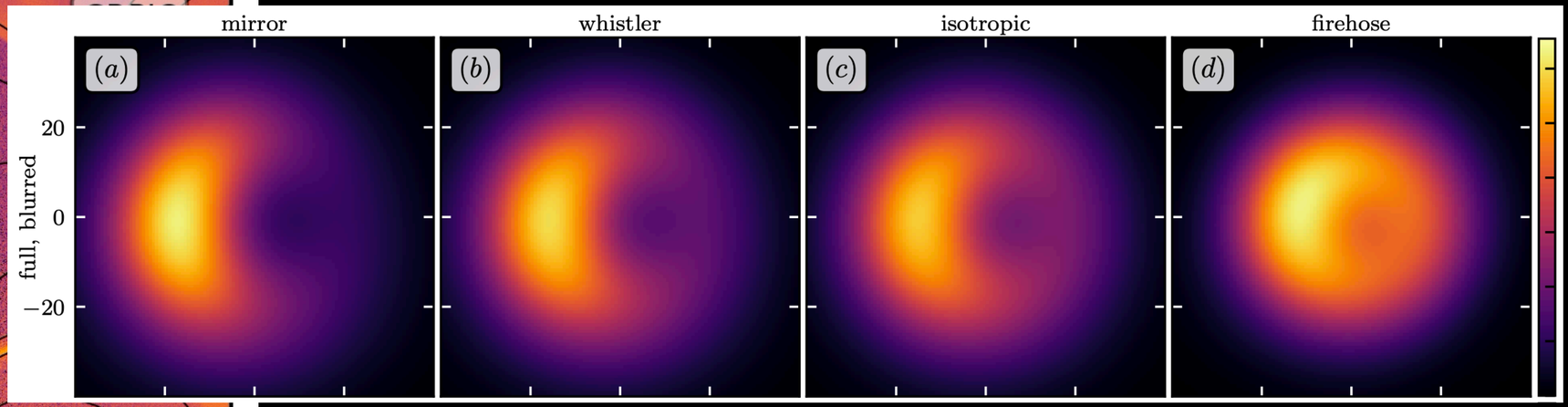


Galishnikova+23 (Parfrey+19, Vos+24)

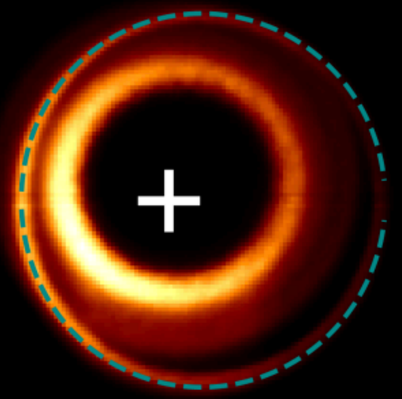


Theoretical Developments….





Incorporating kinetic effects from particle-in-cell simulations and temperature anisotropies



Crinquand+23, Galishnikova+23

AAS245

Summary

- Moving towards a statistical regime with better constraints for M/D for M87

- Testing black hole accretion theory:
  magnetic field structure, image variability and turbulence

- Lots to look forward to!
  Movies, Polarization, Disk-Jet connection

- Tests of extreme plasma physics for the broader community: radiation, alternate black hole feeding modes, kinetic simulations



Event Horizon Telescope